

ICCT MIS — Management Information System

Defense Reviewer • Resource Management & Borrowing System
Live Demo: <https://icct-mis.classapparelph.com/>

1. ANO BA ANG SYSTEM NA ITO?

Isang **Resource Management & Borrowing System** para sa ICCT Colleges — parang library borrowing system, pero para sa school equipment tulad ng HDMI cables, projector, laptop, speaker, at iba pa.

Two Types of Users

Role	Puwedeng Gawin
Admin	Mag-add ng equipment, approve/reject borrow requests, mark as returned, manage users
Faculty	Browse available equipment, mag-request na manghiram, tingnan history ng borrows

Real-life scenario: Gusto ng prof manghiram ng projector para sa class niya → login as Faculty → browse equipment → request borrow → hintay ng Admin approval → kunin ang projector → gamitin → ibalik → admin marks as returned.

2. TECH STACK (Simple Explanation)

Backend — Laravel (PHP)

"Yung utak ng system. Siyang nagpo-process ng login, nagse-save ng data sa database, nagde-decide kung approved or rejected ang request."

Database — MySQL

"Yung storage. Dito naka-save ang users, equipment, borrow requests, at notifications."

Frontend — Tailwind CSS + Blade Templates

"Yung itsura ng website. Yung magagandang colors, rounded corners, at responsive layout — yun ang Tailwind. Yung Blade templates ang paraan ng Laravel para mag-display ng HTML."

Authentication — Laravel Breeze

"Yung sistema ng login at register. Hindi makakapasok ang user kung hindi naka-log in."

3. DATABASE TABLES (Madalas itanong sa defense)

Table	Nilalaman
users	Admin at Faculty accounts (name, email, password, role)
resources	List ng equipment (name, quantity, description)
borrows	Borrow requests (sinong user, anong resource, ilan, status, dates)
notifications	Notifications para sa users (approved/rejected/returned alerts)

Relationships: One User → many Borrows. One Resource → many Borrows. Ang Borrow ang *linking table* na nagkokonekta ng User at Resource.

4. MAIN FEATURES AT PAANO GUMAGANA

4.1 Authentication System

- User nagre-register → may email verification
- Login with role-based redirect (Admin → admin dashboard, Faculty → faculty dashboard)
- May "Forgot Password" at OTP verification

4.2 Resource Management (Admin only)

- Admin puwedeng mag-add, edit, delete ng equipment
- Auto-compute ng available quantity (total - currently borrowed)

4.3 Borrowing Workflow (Ang Puso ng System)

```
Faculty requests → status: PENDING
↓
Admin approves or rejects
↓
Kung APPROVED → puwedeng kunin ang equipment
↓
Admin marks as RETURNED pag naibalik
↓
Complete!
```

4.4 Notifications

- Auto-notify kapag approved, rejected, o returned
- May unread count badge sa navigation
- Puwedeng mark as read o clear all

4.5 Export to CSV

- Admin puwedeng mag-download ng borrow records bilang CSV file (Excel-compatible)

5. COMMON DEFENSE QUESTIONS + SAGOT

Q: Bakit Laravel ang ginamit at hindi pure HTML/CSS/JS?

Sagot: "Gumamit kami ng Laravel dahil mas organized ang code. May MVC pattern — Model (database), View (design), Controller (logic) — hiwalay-hiwalay, hindi magulo. Plus built-in na ang authentication at security features, kaya mas safe at mas mabilis gumawa ng web application."

Q: Anong pinaka-challenging na part sa project?

Sagot: "Yung borrowing workflow — ang status transitions (pending → approved → borrowed → returned), at ang auto-compute ng available quantity para hindi ma-overborrow ang equipment. Kailangan accurate lagi ang quantity computation."

Q: Paano ninyo sinigurado na hindi lalagpas sa available quantity ang borrow?

Sagot: "Sa approve function, chine-check muna namin kung ilan pa ang available. Pag kulang, mag-e-error at hindi magpo-proceed ang approval."

Q: Responsive ba sa mobile?

Sagot: "Opo. Naka-Tailwind CSS po kaya automatic na nag-a-adjust sa screen size. Sa mobile, nagiging card-based layout ang tables imbes na mag-scroll horizontally."

Q: Anong security features ang meron?

Sagot: "May authentication (login/register), role-based access (admin lang puwedeng mag-approve), email verification, at session management. Hindi basta makakapasok ang hindi naka-log in."

Q: Paano ang database design — bakit ganyan ang relationship?

Sagot: "One-to-many po. Isang user puwedeng magkaroon ng maraming borrows. Isang resource puwedeng i-borrow ng maraming beses. Ginawa naming ganito para efficient ang queries at iwas redundant data."

Q: Paano gumagana ang notification system?

Sagot: "Kapag may nag-approve o nag-reject, automatic po kaming nagce-create ng notification record sa database. Pag nag-load ng dashboard, chine-check kung may unread notifications at nagdi-display ng badge."

Q: Ilang users ang kayang i-handle ng system?

Sagot: "Gamit ang Laravel at MySQL, kaya po ng daan-daan hanggang libo-libong users depende sa server capacity. Naka-deploy po ito sa DigitalOcean cloud server na may 4GB RAM."

6. CODE SNIPPETS NA DAPAT ALAM NYO

6.1 Ang Borrow Approval Logic (Admin/BorrowController.php)

```
public function approve(Borrow $borrow)
{
    $resource = $borrow->resource;
    $available = $resource->availableQuantity();

    if ($borrow->quantity > $available) {
        return back()->with('error', 'Not enough available quantity.');
```

}

```
    $borrow->update(['status' => 'approved', 'borrowed_at' => now()]);

    Notification::create([
        'user_id' => $borrow->user_id,
        'title' => 'Borrow Request Approved',
        'message' => "Your request for {$borrow->quantity}x {$resource->name}
                    has been approved.",
    ]);

    return back()->with('success', 'Borrow request approved.');
```

}

6.2 Paano mag-compute ng available quantity (Resource.php Model)

```
public function availableQuantity()
{
    $borrowed = $this->borrows()
        ->whereIn('status', ['approved', 'borrowed'])
        ->sum('quantity');
    return $this->quantity - $borrowed;
}
```

6.3 Routes — Paano naka-setup ang pages (web.php)

```
// Admin routes — naka-protect ng 'role:admin' middleware
Route::middleware('role:admin')->prefix('admin')->group(function () {
    Route::resource('resources', AdminResourceController::class);
    Route::get('/borrows/pending', [AdminBorrowController::class, 'pending']);
    Route::post('/borrows/{borrow}/approve',
        [AdminBorrowController::class, 'approve']);
});

// Faculty routes — naka-protect ng 'role:faculty' middleware
Route::middleware('role:faculty')->prefix('faculty')->group(function () {
    Route::get('/resources', [FacultyResourceController::class, 'index']);
    Route::post('/borrows', [FacultyBorrowController::class, 'store']);
});
```

6.4 Database Migration — borrows table

```
Schema::create('borrows', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained()->onDelete('cascade');
    $table->foreignId('resource_id')->constrained()->onDelete('cascade');
    $table->integer('quantity');
    $table->enum('status', ['pending', 'approved', 'rejected',
        'borrowed', 'returned'])->default('pending');
    $table->timestamp('borrowed_at')->nullable();
    $table->timestamp('due_at')->nullable();
    $table->timestamp('returned_at')->nullable();
    $table->text('notes')->nullable();
    $table->timestamps();
});
```

7. CREDENTIALS FOR DEMO



Role	Email	Password
Admin	admin@icct.edu	admin123
Faculty	faculty@icct.edu	faculty123

Sample Equipment: HDMI Cable, Remote Control, Extension Wire, Speaker, Projector, Laptop

8. POSSIBLE ENHANCEMENTS (Future Add-ons)

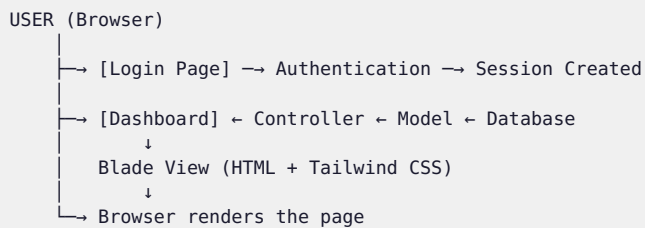
- QR code scanning for equipment check-in/check-out
- Email/SMS reminders for overdue returns
- Dashboard analytics (most borrowed items, usage trends)
- Approval via email link (approve without logging in)
- Audit logs (sinong admin nag-approve, anong oras)
- Print-ready borrowing receipt

9. GLOSSARY OF TERMS

Term	Meaning
MVC	Model-View-Controller — pattern na naghihiwalay ng data (Model), design (View), at logic (Controller)
Migration	Code na gumagawa o nagmi-modify ng database tables
Seeder	Code na naglalagay ng sample data sa database
Middleware	"Bouncer" — chine-check kung allowed ang user na pumunta sa isang page
Blade	Templating engine ng Laravel — paraan para magsulat ng HTML na may PHP
Route	URL address na naka-link sa isang page o function
Model	Kumakatawan sa isang database table sa code
Controller	Nagha-handle ng requests at nagre-return ng response
Foreign Key	Link sa pagitan ng dalawang tables (e.g. user_id sa borrows → users table)
CSV Export	Pag-download ng data sa format na nababasa ng Excel
Composer	Package manager ng PHP — nagda-download ng libraries tulad ng Laravel
Vite	Build tool — nagco-compile ng CSS/JS assets para sa production

10. SYSTEM ARCHITECTURE

Simplified na diagram kung paano dumadaloy ang data:



Flow ng Borrow Request:

```
Faculty clicks "Borrow" → Route → FacultyBorrowController@store
→ validates input → saves to borrows table (status: pending)
→ redirects back with success message

Admin sees pending → clicks "Approve" → Route
→ AdminBorrowController@approve
→ checks availableQuantity()
→ updates borrows table (status: approved)
→ creates notification record
→ Faculty sees notification on dashboard
```